

Research Article

In silico study of cancer stage-specific DNA methylation pattern in White breast cancer patients based on TCGA dataset

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ABSTRACT

Background: Breast cancer is one of the most common types of cancer among women. As current breast cancer treatments are still ineffective, we assess the methylation pattern of White breast cancer patients across cancer stage based on The Cancer Genome Atlas (TCGA) dataset. Significant hypermethylation and hypomethylation can regulate the gene expression, thus becoming potential biomarkers in breast cancer tumorigenesis.

Methods: DNA methylation data was downloaded using TCGA Assembler 2 based on race-specific metadata of TCGA - Breast Invasive Carcinoma (TCGA-BRCA) project from Genomic Data Commons (GDC) Data Portal. After the data was divided into each cancer stage, duplicated data of each patient was removed using OMICSBind, while differentially-expressed probes were identified using edgeR. The resulting probes were validated based on correlation and regression analysis with the gene expression, ANOVA between cancer stages, ROC curve per stage, as well as databases.

Results: Based on the White dataset, we found 66 significant hypermethylated genes with logFC > 1.8 between Stage I-III. From this number, three epigenetic-regulated, stage-specific genes are proposed to be the detection biomarkers of breast cancer due to significant aberrant gene expression and/or low mutation ratio among breast cancer patients: ABCC9 (Stage III), SHISA3 (Stage II), and POU4F1 (Stage I-II).

Conclusions: Our study shows that ABCC9, SHISA3, and POU4F1 are potential stage-specific detection biomarkers of breast cancer for White individuals, whereas their roles in other races need to be studied further.

1. Introduction

Breast cancer is the most common cancer among women. It is a heterogeneous disease with multiple clinical, histopathologic, and molecular subtypes, exhibiting different therapeutic responses and prognoses (Jagtap, 2018). In Indonesia, breast cancer had the highest incidence and prevalence compared to other types of cancer (Global Cancer Observatory, 2019). It was also the second leading cause of cancer-related death after lung cancer, with a mortality rate of 11 % (Global Cancer Observatory, 2019). Worldwide, breast cancer accounted for 11.6 % new occurrence of cancer in 2008, with 15 % of mortality rate among females (Bray et al., 2018). These highlight the importance

of understanding the regulation of cancer. Up until now, there are three main treatments for cancer: radiotherapy, surgery, and chemotherapy. Unfortunately, all of them are still not effective to completely stop the breast cancer growth and/or remove it (Chakraborty and Rahman, 2012). In this case, we proposed a new perspective in developing breast cancer treatment: epigenetic regulation.

Epigenetic regulation is the study of heritable changes in gene expression which is not caused by mutations in the gene sequence, including DNA methylation (Jaenisch and Bird, 2003). DNA methylation itself is the attachment of methyl (CH₃) group to DNA molecule, typically cytosine, and thus often modifying the function of the genes and affecting the gene expression (Jin et al., 2011). Most of the time,

Abbreviations: TCGA, The Cancer Genome Atlas; BRCA, breast invasive carcinoma; GDC Data Portal, Genomic Data Commons Data Portal; ANOVA, analysis of variance; ROC, receiver operating characteristic curve; DNA, deoxyribonucleic acid; QL-GLM, quasi-likelihood - generalized linear model; Hg38, human genome assembly Genome Reference Consortium Human Build 38; AUC, area under curve; EWAS, epigenome-wide association studies; COSMIC, Catalogue of Somatic Mutations in Cancer; StringDB, Search Tool for the Retrieval of Interacting proteins database; logFC, log fold-change; FDR, false discovery rate; miRNA, micro-ribonucleic acid; HCC, hepatocellular carcinoma; OSCC, oral squamous cell carcinoma; N/A, not applicable; ER, estrogen receptor; ERE, estrogen response element; ATP, adenosine triphosphate.

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Table 1
Initial number of patients based on race and cancer stage.

Cancer Stage	African-American or Black	White
0 (healthy)	4	90
I	26	95
II	56	171
III	31	156
IV	5	5
N/A	42	150

methylation inhibits the expression of the target gene as it prevents transcription and thus translation of the gene. (Kulis and Esteller, 2010) further showed that DNA methylation plays a significant role in tumorigenesis. In this study, we aim to propose novel methylation-regulated gene biomarkers whose role in breast cancer development is still poorly-studied. As we inferred the role of the genes based on their expression and methylation level (instead of their known functions), our biomarkers will be limited for breast cancer detection until further validation. We will utilize The Cancer Genome Atlas (TCGA) dataset, which is a widescale cancer genomics program that sequences and compares the genetic difference between cancer patients and normal samples (NIH, 2021a). By looking at the methylation coordinate and level of each probe, we could map the location of the probe to its respective gene and predict its role in breast cancer development. While the association between gene expression and methylation level is not always linear (e.g. hypermethylated genes, particularly ones that happen in gene body, might have high expression and act as oncogene (e.g. (Su et al., 2018; Arechederra et al., 2018)), here we only focus on the common occurrence of direct, negative relationship between the two factors (i.e.. hypermethylation will inhibit the expression of the gene, and vice versa). This can shed an idea of the genes' role in breast cancer development (oncogene or tumor suppressor gene (Muir and Nunney, 2015; Herman, 1999)).

In fact, DNA methylation patterns in breast cancer have been widely studied via the incorporation of other variables (Feng and Jin, 2018; Chen et al., 2019; Zhang and Cui, 2019), including race (Ambrosone et al., 2014) and cancer stage (McMahon et al., 2017). However, there is still a lack of study that incorporates multiple variables at once. In this study, we will divide the dataset based on two variables: race and cancer stage. Both were chosen by considering the specificity of epigenetic regulation (Chen et al., 2013; Coit et al., 2015) and the data availability in TCGA. In the next step, the methylation (beta) value for each race/-stage group will be compared to the healthy samples. Beta value measures the methylation level of a gene by calculating the proportion of methylation at a given locus (Wessely and Emes, 2012). Hypermethylated and hypomethylated probes will be mapped to their genes, while their roles in breast cancer will be validated by looking at the correlation and causation with the target gene's expression, diagnostic ability to differentiate healthy and cancer patients based on receiver operating characteristic curve, mean differences between cancer stages based on analysis of variance, and databases. This study aims to identify specific methylation-regulated genes for each cancer stage, providing a basis for further development of epigenetics-based breast cancer treatment or therapy for specific race and cancer stages.

2. Results

Initial race-based metadata resulted in 164 patients for African-American/Black dataset (160 cancer patients, 4 normal) and 667 for White dataset (577 cancer patients, 90 normal). After the dataset was filtered, samples with unidentified cancer stage (i.e. N/A; Table 1) were removed from the analysis. Based on Table 1, there are only five patient's data for Stage IV cancer (each race) and four normal samples for African-American/Black race. After the methylation data was filtered based on the logFC, *p* value, and FDR, the number of probes reduced

Table 2
Number of hypermethylated-probes with different filtering criteria.

Cancer Stage	African-American or Black			White		
	logFC	<i>p</i> , FDR	logFC, <i>p</i> , FDR	logFC	<i>p</i> , FDR	logFC, <i>p</i> , FDR
I	449	0	0	268	5170	268
II	1118	0	0	541	13,816	541
III	1122	0	0	870	15,176	870
IV	2099	0	0	1373	0	0

*logFC > 1.5; *p* < 0.05; FDR < 0.05.

Table 3
Number of hypomethylated-probes with different filtering criteria.

Cancer Stage	African-American or Black			White		
	logFC	<i>p</i> , FDR	logFC, <i>p</i> , FDR	logFC	<i>p</i> , FDR	logFC, <i>p</i> , FDR
I	0	0	0	0	220	0
II	2	0	0	1	2330	1
III	0	0	0	0	2347	0
IV	2	0	0	0	0	0

*logFC < -1.5; *p* < 0.05; FDR < 0.05.

drastically, particularly for African-American/Black dataset where none of the probes passed the criteria (Tables 2 and 3). On the other hand, the White dataset returned more than 1500 hits for the hypermethylated probes for cancer Stage I-III (Table 2), but not the others (Table 3). Therefore, we will focus on the results from hypermethylated probes in White dataset, Stage I, II, and III.

Interestingly, there is only one methylation-regulated microRNA (miRNA), miR-129-2, even though the reference file contains 1007 miRNA entries. As there are more than 2000 human miRNA that have been identified Alles et al. (2019), further study is required to assess the correlation between miRNA regulation and DNA methylation. In this study, we limit our analysis to probes with identified target gene/miRNA according to the TCGA DNA methylation reference file. As there are 268, 541, and 780 hypermethylated probes among White patients in Stage I, II, and III, respectively (Table 2), we increased the threshold into logFC > 1.8. This resulted in six, sixteen, and sixty-two hypermethylated genes for Stage I, II, and III, respectively. Then, we further divided the genes based on their occurrence or absence in each stage (Table 4). Due to the new threshold, there is no hypermethylated gene that is specific for Stage I. Their methylation patterns were further plotted by using R Studio (Fig. 1). Overall, the cancer stage with the highest average of methylation is Stage III (Fig. 1), which might imply that these genes become more suppressed as the breast cancer stage escalates. Genes with the highest level of hypermethylation (beta-value) are as follows (Healthy/Stage I/Stage II/Stage III): ADAMTS20 (0.066/0.523/0.556/0.572), USP44 (0.083/0.582/0.629/0.656), ENPP2 (0.075/0.540/0.578/0.605), RYR2 (0.050/0.531/0.541/0.589), and CYP26A1 (0.064/0.453/0.542/0.559). All of them are statistically hypermethylated in at least two stages of cancer (Table 4) and have been previously reported to be correlated with breast cancer (Kumar et al., 2012; Guo et al., 2018; Lan et al., 2016; Abdul et al., 2008; Cha and Koo, 2019; Osanai and Lee, 2015).

From Table 4, there are genes whose correlations with breast cancer in specific cancer stage have not been well-studied (based on literature review): TMEM145 (all stages); DBX1 and SLC4A10 (stage II); ABCC9, C14orf39, C17orf64, GCM2, GRM6, ILDR2, POU4F1, PP12613, SHISA3, TMEM155, and TRIM67 (stage III). According to unpublished study from Nasution et al., there are two out of 14 genes with suppressed expression among breast cancer patients: ABCC9 and SHISA3 (Table 5). In contrast, other genes (TMEM145, SLC4A10, ILDR2, POU4F1, TMEM155, TRIM67) are overexpressed despite their hypermethylated state. As mentioned before, this might infer the presence of multiple

Table 4Stage-specific hypermethylated genes based on significantly methylated probes in White dataset. Threshold: $\log_{2}FC > 1.8$, $p < 0.05$, FDR < 0.05 .

Stage I-III	Stage II-III	Stage II	Stage III			
CCDC181	ADAMTS20	AC009487.5	ABCC9	DNM3	LAMA1	RP11-175K6.1
HSD17B12	CYP26A1	DBX1	AHRR	EBF1	LEPR	RP11-445F12.1
MIR129-2	DOCK2	SLC4A10	BCAT1	FOXC2	LHX1	RP11-521O16.1
PRKCB	ENPP2	ZNF300P1	BHLHE23	GCM2	NID2	RP11-521O16.2
RYR2	T		C14orf39	GRM6	NPY	RP11-567J20.2
TMEM145	USP44		C17orf64	HOXD-AS2	PCDH8	RP11-637O19.2
			CD38	HOXD8	POU4F1	RP11-637O19.3
			CDO1	ILDR2	PP12613	RP11-662I13.3
			CHST2	IRF4	PROX1	RP11-770E5.1
			DNER	ITPKA	PROX1-AS1	RP11-96H19.1
			SALL2	SHISA3	SIX6	SLC38A4
			SNCA	SNCA-AS1	TMEM155	TRIM67
			USP32	VWC2		

methylation probes in the gene regulation.

In order to limit the genes into the ones that are epigenetically regulated, we further compared the percentage between mutation and hyper-/hypomethylation occurrences of the genes in breast cancer patients based on the COSMIC database (Table 6). According to the table, there are three genes with significantly higher hypermethylation occurrence compared to mutation: DBX1 (18.95 %), POU4F1 (8.49 %), and TRIM67 (11.17 %). All three of them are mainly expressed in the brain cells, playing a role during the development of the central nervous system [Uniprot ID: A6NMT0 (DBX1), Q01851 (POU4F1), Q6ZTA4 (TRIM67)]. In addition, there was no entry for PP12613, as it encodes for long non-coding RNA (lncRNA) instead of protein. While we cannot completely overlook these genes, additional validation steps (either experimental or computational) are needed to assess their roles in breast cancer tumorigenesis.

There are five epigenetic-regulated genes (with low mutation level) that might become novel breast cancer biomarkers, with their methylation values shown in bracket (Normal/Stage I/Stage II/Stage III): ABCC9 (0.070/0.523/0.542/0.585), SHISA3 (0.042/0.383/0.389/0.467), DBX1 (0.047/0.447/0.513/0.477), POU4F1 (0.032/0.360/0.385/0.449), and TRIM67 (0.037/0.353/0.401/0.454). This is supported by the EWAS Atlas database, where all of the probes were found hypermethylated in several cases of cancer, such as hepatocellular carcinoma (HCC) and oral squamous cell carcinoma (OSCC). Hypermethylation in DBX1, in fact, has been associated with breast cancer from other TCGA-based study (de Almeida et al., 2019). Their methylation locations (based on TCGA reference file and EWAS Atlas) are shown in Table 7. Even though the overall methylation level of each gene is low to moderate (from the scale 0.0–1.0; Table 8), they are clearly hypermethylated among breast cancer patients with significant occurrence in the promoters (initiation region for transcription, located upstream to the transcription start site) which have been associated with the repression of transcription activity (Moore et al., 2012). Moreover, the methylation value between cancer stage is statically different for SHISA3 ($p = 0.036$), POU4F1 ($p = 0.024$), and TRIM67 ($p = 0.018$). This might infer the potential role of these genes as progressive marker for breast cancer tumorigenesis.

Pearson correlation and linear regression model (LGM) analysis were done to ensure the negative correlation and directionality between methylation probes (independent variable) and the gene expression (dependent variable; Nasution et al., unpublished). According to Table 9, significant (p -value < 0.05) negative correlations are found in three genes: ABCC9 (Stage III), SHISA3 (Stage II), and POU4F1 (Stage I-II). The directionality is confirmed by the result from LGM (Table 10),

where significant negative regression coefficients are found in those pairs with significant negative correlation (Table 9). Moreover, the ROC curves infer good sensitivity and specificity for each gene in differentiating the healthy and cancer patients in each cancer stage (Fig. 2), with >80 % AUC (except for SHISA3 in Stage II; Table 11). These results support the potential of these genes as detection, even diagnostic stage-specific biomarker.

3. Discussion

In this study, we found no significant results for African-American/Black race due to the limited number of healthy data. This significantly affected the value of p and FDR. For the White data set, the number of hypomethylated probes with $\log_{2}FC < -0.5$ was only one (which might infer a lack of hypomethylation events in breast cancer). Thus, we retrieved the top-five hypomethylated probes for Stage I-III (White dataset, not shown). By selecting the ones with known gene targets, each stage returned the same hypomethylated genes: CLTC and LEFTY1. CLTC (clathrin heavy chain) is a major protein component in intracellular organelles; it is involved in intracellular trafficking of receptors and endocytosis. Previous study (Jiao et al., 2013) showed that it was overexpressed in breast cancer patients, confirming its hypomethylated state among White breast cancer patients. On the contrary, LEFTY1 (left-right determination factor 1) was found to be negatively-correlated with the aggressiveness of malignant breast cancer cells (Kalyan et al., 2017). The contradiction between the down-regulated expression of the gene and its hypomethylation state might infer that the gene expression is regulated by more than one methylation site.

After several validation steps, three genes become candidates as novel biomarkers for breast cancer (White dataset): ABCC9 (Stage III), SHISA3 (Stage II), and POU4F1 (Stage I-II). Even though the correlation coefficient for each pair was not strong (<0.4 ; Table 9), we cannot overlook these interactions as: (i) the correlation and directionality are significant (ii) the expression of one gene might be controlled by multiple methylation sites, and (iii) data limitation might weaken (or strengthen) the correlation coefficient. ABCC9 (ATP Binding Cassette Subfamily C Member 9) is a member of the ATP-binding cassette (ABC) transporters protein family (Uniprot ID: O60706). It plays a role in the active transport of various substrates, such as sugars, proteins, and lipids, by interacting with other proteins, such as ABCA9 and ABCA8 (Fig. 3A). ABC transporters themselves have been found to cause drug resistance among cancer patients by effluxing the drug out of the cells and thus reducing its bioavailability (Choi, 2005). On the other hand,

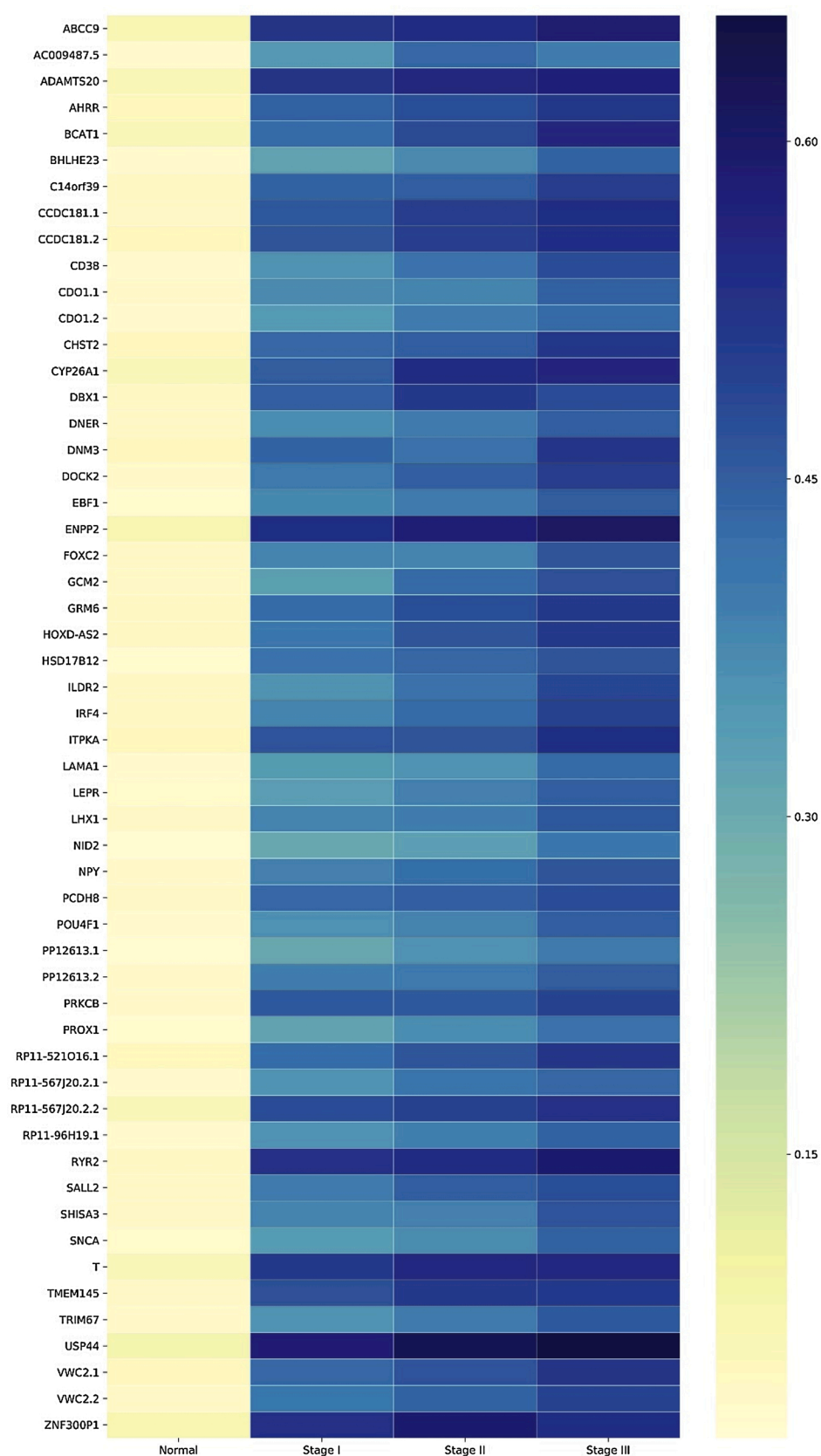


Fig. 1. Methylation levels (β -value) based on significantly methylated probes in White dataset.

Table 5

Hypermethylated and differentially expressed genes based on significantly methylated probes in White dataset. Methylation data derived from this study, while gene expression data derived from Nasution et al. (unpublished). Threshold: $p < 0.05$, FDR < 0.05 .

Genes	Methylation (logFC)			Expression (logFC)		
	Stage I	Stage II	Stage III	Stage I	Stage II	Stage III
TMEM145	1.828	1.934	1.922	4.362	5.198	4.991
DBX1	N/A	1.884	N/A	N/A	N/A	N/A
SLC4A10	N/A	1.829	N/A	2.435	2.928	3.006
ABCC9	N/A	N/A	1.859	-2.232	-2.121	-2.064
C14orf39	N/A	N/A	1.828	N/A	N/A	N/A
C17orf64	N/A	N/A	1.808	N/A	N/A	N/A
GCM2	N/A	N/A	1.803	N/A	N/A	N/A
GRM6	N/A	N/A	1.874	N/A	N/A	N/A
ILDR2	N/A	N/A	1.811	2.170	2.080	1.986
POU4F1	N/A	N/A	1.861	3.002	3.201	N/A
PP12613	N/A	N/A	1.821	N/A	N/A	N/A
SHISA3	N/A	N/A	1.821	-2.579	-2.587	-2.320
TMEM155	N/A	N/A	1.810	0.976	1.045	1.030
TRIM67	N/A	N/A	1.832	3.302	N/A	1.561

Table 6

Validation based on COSMIC database.

Genes	% Samples with Mutation	% Samples with Gene Overexpression	% Samples with Hyper ^R / Hypomethylation ^O
TMEM145	0.82	2.72	N/A
DBX1	0.50	3.71	18.95 ^R
SLC4A10	8.93	2.08	-
ABCC9	5.71	3.35	1.70
C14orf39	1.90	3.89	0.42 ^R
C17orf64	0.39	5.34	N/A
GCM2	0.50	2.63	2.55 ^R
GRM6	1.18	3.53	2.69 ^R , 2.400
ILDR2	3.11	3.44	0.14 ^R , 0.850
POU4F1	0.35	8.97	8.49 ^R
PP12613	N/A	N/A	N/A
SHISA3	0.20	2.36	3.540
TMEM155	0.35	1.99	0.71 ^R , 3.250
TRIM67	2.29	1.72	11.17 ^R

Table 7

Methylation locations of five potential epigenetic-regulated genes in breast cancer. Island: >500bp CpG-rich region; N_Shelf: 2-4 kb upstream from the CpG Island.

Probes	Gene	Location	Location to CpG Island
cg27555582	ABCC9	Promoter	Island
cg14866200	SHISA3	Promoter	Island
cg13445796	DBX1	Promoter	Island
cg27398263	POU4F1	Promoter	N_Shelf
cg27504802	TRIM67	Promoter	Island

Table 8

Average methylation levels (β -value) based on all probes in White dataset.

Genes	Normal	Stage I	Stage II	Stage III
ABCC9	0.321	0.491	0.509	0.535
SHISA3	0.264	0.344	0.353	0.379
DBX1	0.154	0.344	0.357	0.354
POU4F1	0.091	0.337	0.355	0.394
TRIM67	0.248	0.375	0.400	0.413

Table 9

Pearson Correlation between methylation and gene expression in White breast cancer patients. Value: correlation coefficient (p value). Bold: significant correlation (p value < 0.05).

Genes	Stage I	Stage II	Stage III
cg27555582 - ABCC9	-0.180 (0.082)	-0.094 (0.318)	-0.188 (0.019)
cg14866200 - SHISA3	-0.096 (0.359)	-0.206 (0.028)	-0.095 (0.237)
cg13445796 - DBX1	0.050 (0.633)	-0.093 (0.327)	-0.016 (0.848)
cg27398263 - POU4F1	-0.249 (0.016)	-0.372 (<0.001)	-0.126 (0.116)
cg27504802 - TRIM67	-0.126 (0.228)	0.037 (0.700)	-0.142 (0.077)

Table 10

Linear Regression between methylation and gene expression in White breast cancer patients. Value: regression coefficient (p value). Bold: significant correlation (p value < 0.05).

Genes	Stage I	Stage II	Stage III
cg27555582 - ABCC9	-445.6 (0.082)	-193.0 (0.318)	-444.9 (0.019)
cg14866200 - SHISA3	-8.215 (0.359)	-24.125 (0.028)	-6.474 (0.237)
cg13445796 - DBX1	0.045 (0.633)	-0.739 (0.327)	-0.018 (0.848)
cg27398263 - POU4F1	-88.36 (0.016)	-87.92 (<0.001)	-17.341 (0.116)
cg27504802 - TRIM67	-722.5 (0.228)	2.926 (0.699)	-91.31 (0.077)

SHISA3 (Shisa Family Member 3) plays a role in maturation of pre-somitic mesoderm cells (Uniprot ID: A0PJX4). Even though there is still a limited information of its interaction in tumorigenesis, SHISA3 has been found to be hypermethylated among breast cancer patients and become a potential tumor suppressor gene for breast cancer (Shahzad et al., 2020). Lastly, POU4F1 (POU Domain, Class 4, Transcription Factor 1) is a member of neural transcriptions factors (Uniprot ID: Q01851). It interacts with tumor protein 53 (p53) and estrogen receptor (Fig. 3B) which is found to inhibit the transcription of proapoptotic target genes (Hudson et al., 2004) and increase the binding of estrogen receptor (ER) to estrogen response element (ERE) (Budhram-Mahadeo et al., 1998), respectively. The hypermethylated state of each gene in breast cancer patients has been found before (e.g. (Zhang et al., 2020; Muriithi et al., 2020; Cai et al., 2016)); however, none considered the specificity of epigenetic regulation in different cancer stage. Our results show that hypermethylation of each gene is only significant in specific cancer stage. The results and future prospects of this study are summarized in Fig. 4, where further studies are the ones that we plan to do in the near future to confirm our results.

4. Conclusions

Our study shows that there are cancer stage-specific epigenetic (DNA methylation) regulations in breast cancer. They are denoted by high logFC differences between cancer and normal individuals, particularly in Stage I, II, and III among White individuals. Here, we proposed three potential cancer stage-specific hypermethylated biomarkers for breast cancer: ABCC9 (Stage III), SHISA3 (Stage II), and POU4F1 (Stage I-II) due to their aberrant expressions and/or low mutation incidence between breast cancer patients. In particular, their respective methylation probes performed well in breast cancer detection, while SHISA3 and POU4F1 also had statistically different methylation patterns across stages. While these results are derived from White dataset and currently limited for cancer detection, further experimental study is required to validate the specificity of those genes in breast cancer tumorigenesis among other races as diagnostic, prognostic, or even progressive

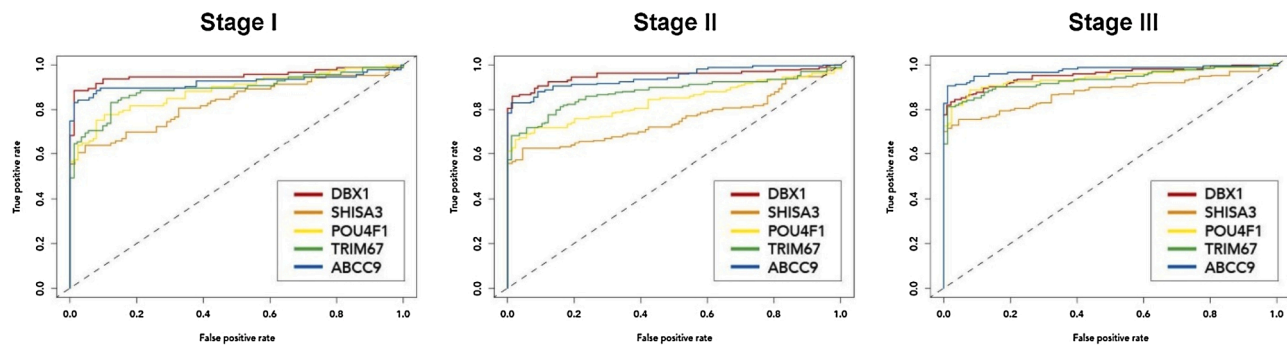


Fig. 2. ROC of five potential epigenetic-regulated genes in each breast cancer stage.

Table 11
Percentage of AUC from the ROC plots (Fig. 2). Bold: significant correlation from Table 9.

Genes	Stage I	Stage II	Stage III
cg27555582 - ABCC9	92.19	94.49	97.52
cg14866200 - SHISA3	83.45	75.85	88.03
cg13445796 - DBX1	95.25	95.43	95.36
cg27398263 - POU4F1	88.73	83.89	94.16
cg27504802 - TRIM67	89.12	88.01	93.29

biomarkers.

5. Materials and methods

5.1. Dataset

In this study, our dataset is normalized, aggregated level 3 of TCGA-BRCA (The Cancer Genome Atlas - Breast Invasive Carcinoma), which is

already free from batch effects (NIH, 2021b). The data is available at Genomic Data Commons (GDC) Data Portal (Grossman et al., 2016).

5.2. Data pre-processing

First, the metadata and clinical data of the breast cancer patients were downloaded based on race (i.e. White or African-American/Black) from GDC Data Portal in October 2019. Other races were excluded from the analysis due to insufficient number of samples when partitioned based on race and cancer stage. Then, the first fifteen characters of each patient's barcode (ID) were retrieved from the metadata by using Python 3.6 (Van Rossum and Drake, 2009), which was subsequently inputted into TCGA Assembler 2.0.6 (Wei et al., 2017) to download the 450k methylation data.

By referring to the clinical data of the breast cancer patients, the dataset for each race was further divided into each cancer stage (i.e. I, II, III, or IV) by using R Studio (RStudio Team, 2015), while the unidentified one (i.e. N/A) was removed from further analysis. This resulted in eight datasets (i.e. four stage cancers each for White and

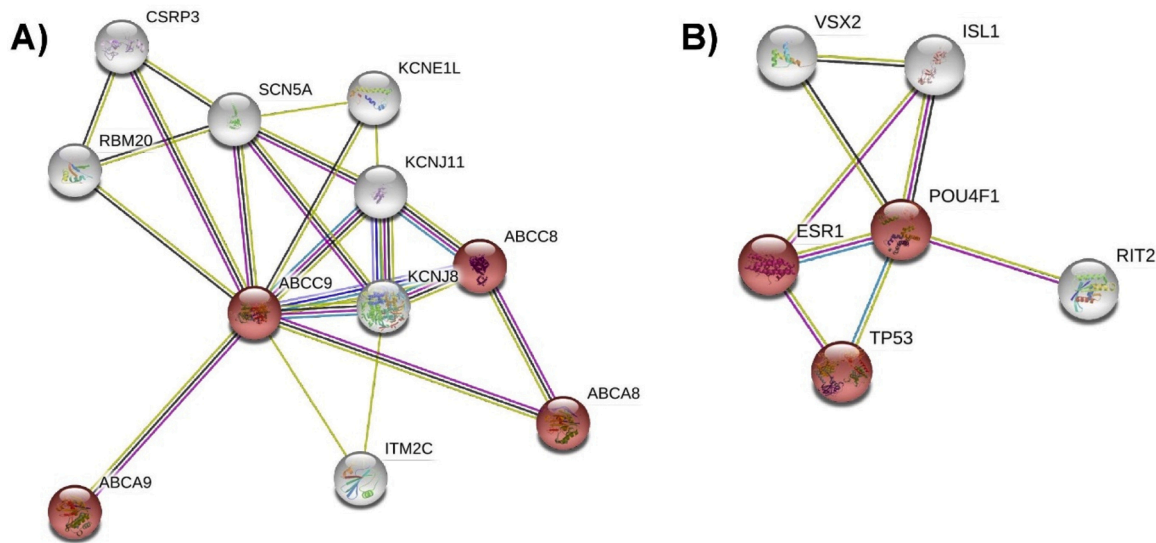


Fig. 3. Protein-protein interaction of ABCC9 (A) and POU4F1 (B). The graph was derived from StringDB (Li et al., 2019). Red color infers interactions mentioned in the previous paragraph.

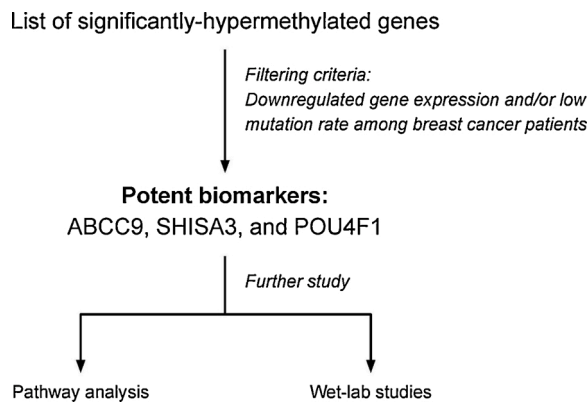


Fig. 4. Result summary of this study. Pathway analysis and wet-lab studies are needed to further confirm our results.

African-American/Black races). However, the data for African-American/Black group were few and thus not analyzed further.

5.3. Identification of hypermethylated and hypomethylated probes

Each dataset was then analyzed by using edgeR (Robinson et al., 2009) to determine the differentially-expressed methylation probes. EdgeR accounts differential expression for count data, performing better compared to other similar software (McCarthy et al., 2012; Zhou et al., 2014). In this case, quasi-likelihood (QL) Generalized Linear Model (GLM) was used to account for dispersion uncertainty between genes (Chen et al., 2020). Then, the result was filtered based on the logFC (logFC > 1.5 for up-regulated probes; logFC < -1.5 for down-regulated probes), p value, and FDR (both must be lower than 0.05). Probes that passed the criteria would be mapped into the gene based on the TCGA DNA methylation reference file that is available in the GDC Data Portal. The file contains the list of probes and their respective target genes. The mapping itself was based on the Hg38 genome annotation.

5.4. Result validation

In order to validate the role of these genes in breast cancer development, we will first assess their expressions among breast cancer patients based on Nasution et al. (unpublished). By conducting similar procedures as our study, they utilized TCGA gene expression data of breast cancer patients to identify race-/stage-specific differentially-expressed genes. As one individual might undergo several tests, which results in multiple non-independent barcodes, the samples from the gene and methylation dataset were matched based on the first 12 characters (comprising project's name, TSS, and participant) by using OMICSBind (Wan et al., 2015). Then, Pearson correlation and linear regression analysis were done between the methylation probe (beta-value) and the expression of its target gene by using R. This step is crucial to ensure that the methylation events lead to promotion (hypomethylation) or inhibition (hypermethylation) of the gene expression. The ability of the genes for detecting breast cancer in each cancer stage was also assessed based on receiver operating characteristic (ROC) curves and its respective area under the curve (AUC) by using ROCR package (Sing et al., 2005). Next, one-way analysis of covariance (ANOVA) was conducted in R Studio to assess the mean methylation differences between cancer stages. We will also validate the genes based on the EWAS Atlas (Li et al., 2019). COSMIC (Catalogue of Somatic Mutations in Cancer) (Tate et al., 2018), and StringDB (Szklarczyk et al., 2018) databases; cancer-related genes whose hyper/hypo-methylation states have been observed and mutations have not been addressed will be chosen as potential epigenetic biomarkers in breast cancer tumorigenesis. The complete procedure was shown in Fig. 5.

Compliance with ethics guidelines

This article does not contain any studies with human participants or animals performed by any of the authors.

Declaration of Competing Interest

The authors report no declarations of interest.

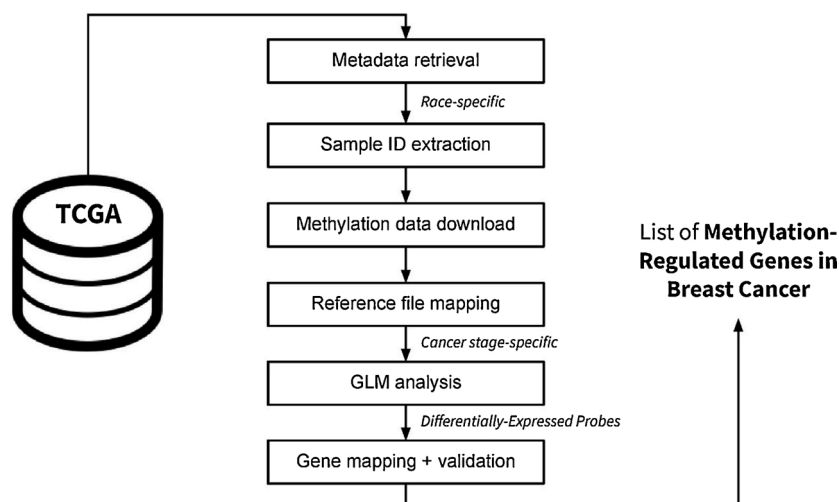


Fig. 5. General procedure of the study.

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